

Readiness to Train

Project Results

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1. Introduction

The goal of this project **was** to develop a **causal framework for optimising training load per player** in professional football, such that each player arrives at match day in the best possible physical condition. The central question: *"Given everything we know about a player this morning, what training intensity should we prescribe today to maximise their match-day performance?"*

This is fundamentally a **causal** question: coaches already make training decisions based on player state, which means the observed data confounds the effect of training with the selection of who receives it. Standard predictive modelling cannot disentangle these; causal methods are required.

The project uses daily monitoring data collected by OH Leuven over approximately 20 months (July 2024 – February 2026), covering **28 first-team players** and **5,235 useful player-day observations**. Four raw datasets were merged into a single analysis-ready file (RTT.xlsx) with 46 engineered features per player-day.

Two auto-generated data dictionaries document every variable: the **Raw Data Dictionary** (*data/raw/Raw Data Dictionary.pdf*) covers all columns across the four raw datasets, while the **RTT Data Dictionary** (*data/processed/RTT Data Dictionary.pdf*) describes all 46 engineered features in the processed dataset, including their temporal positions and derivations.

Three experiments were conducted, each building on the previous. This report tells the story of how we arrived at the results.

2. Repository & Notebooks

The full project is available on GitHub: github.com/andreasgoethals/Readiness-To-Train. It contains the preprocessing pipeline, five ML model implementations (Ridge, Logistic Regression, XGBoost, CatBoost, TabPFN), a causal DAG builder, experiment runner scripts, and all analysis notebooks.

The notebooks are organised at three levels:

Level 0 — Data Quality Checks: *0. Processed_Data_Quality* (automated validation of RTT.xlsx) and *0. TI_Missingness_Analysis* (Training Intensity NaN patterns).

Level 1 — Data Visualisation: *1.1. Match Analysis* (match-level exploration), *1.2. Raw Data Visualisation* (EDA across all 4 raw datasets), *1.3. Processed Data Visualisation* (EDA of RTT.xlsx).

Level 2 — Experiments: *2.1. Experiment 1* (Match Intensity), *2.2. Experiment 2* (Training Intensity), *2.3. Experiment 3* (Status Decrease).

All visualisations are preserved inside the notebooks.

3. Data & Key Variables

Dataset	Rows	Players	Days	Content
Readiness_Data	14,359	28	597	Wellness z-scores, ACWR, GPS benchmarks, medical status
Raw_Data	9,968	84	670	Detailed GPS, heart rate, session metadata
Sessions	1,206	—	670	Match day flags, session types
Games	403	24	216	High-intensity distance/efforts per ball-in-play, minutes

Training Intensity Yesterday (treatment variable): A composite GPS score capturing how hard a player trained. Computed as $\tanh(\text{harmonic_mean}(\text{TD}\%, \text{HSD}\%, \text{Dec}\%, \text{Sprints}\%) / 100)$, where each GPS metric is a percentage of the player's personal match benchmark. Range [0, 1]: 0 = rest, near 1 = match-equivalent effort.

Match Intensity Yesterday (match-day outcome): Geometric mean of high-intensity distance and efforts per ball-in-play minute, scaled by minutes played. Only filled the day after a match.

Status Decrease (short-term outcome): Binary indicator of whether a player's medical status worsened (e.g. Available → Attention). Prevalence ~2%.

Figure 1: Training Intensity per player (sorted by median).

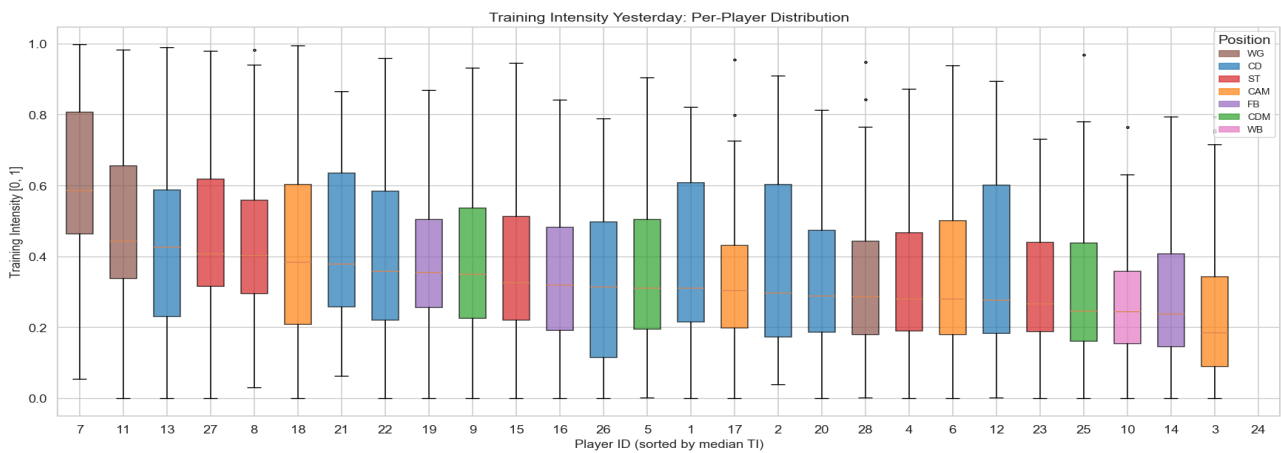
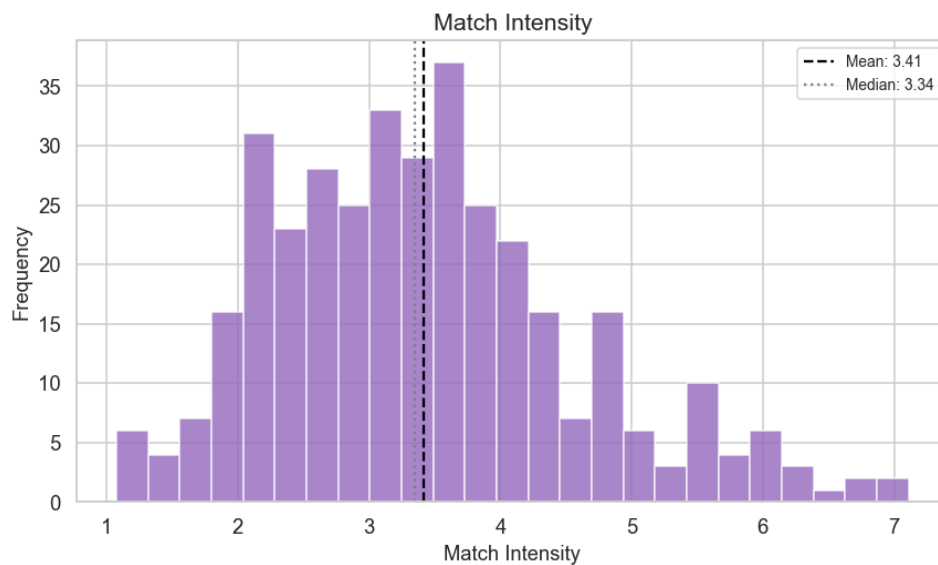


Figure 2: Match Intensity distribution (mean = 3.41).



4. Experiment 1 — Match Intensity Prediction

Before attempting a causal framework, we first needed to verify whether the intended outcome — match-day performance — is even *predictable* from training and wellness data. If standard ML cannot predict it, then causal methods will certainly not identify a causal effect either: causal identification requires that the outcome varies systematically with the treatment *conditional on confounders*, which is impossible if the outcome has no recoverable association with the available covariates in the first place.

Experiment A — Raw Match Intensity

We predicted Match Intensity Yesterday from 13 morning covariates (including Player ID) using Ridge Regression, XGBoost, CatBoost, and TabPFN across 10 lag values. The dataset is small: ~248 training, ~72 test samples (only match-day rows contribute).

Model	Best Lag	RMSE	R2	Pearson r
LinReg	1	1.114	0.266	0.723
TabPFN	10	1.125	0.251	0.633
CatBoost	2	1.194	0.158	0.556
XGBoost	1	1.248	0.080	0.613

LinReg achieves $R^2 = 0.27$ at lag=1. However, the **permutation importance analysis** reveals that **Player ID accounts for almost all the signal** (importance = +0.39, while the next feature is only +0.01). The model is simply learning that some players consistently perform at a higher level than others.

Per-player analysis: All 17 test-set players have negative R^2 except one ($R^2 = 0.50$). The aggregate R^2 of 0.27 is entirely driven by between-player differences, not within-player training effects.

Experiment B — Personal Deviation

To confirm that Player ID was the sole driver, we removed the player fixed effect by predicting **Match Intensity Personal Deviation** (each player's match intensity minus their own expanding-mean baseline). This forces the model to predict *within-player* variation.

Result: **all models achieve R^2 near zero or negative** across all lags. The best result (XGBoost lag=5, $R^2 = -0.001$) is indistinguishable from random. With the player effect removed, there is no recoverable signal linking morning state to match performance.

Conclusion

Match-day performance is NOT predictable from training and wellness data. Experiment A showed apparent signal ($R^2 = 0.27$) driven entirely by Player ID. Experiment B confirmed no within-player signal exists. Since the outcome is not predictable in a non-causal sense, it cannot serve as a causal target. This motivated the pivot to Experiment 2.

5. Experiment 2 — Training Intensity Prediction

Given that match-day performance is unpredictable, we pivoted to a different but equally valuable question: **can we recover the coaching staff's load-assignment decisions from observable morning state?**

If the answer is yes, the fitted model serves as a **data-driven proxy for player Readiness to Train**. The reasoning: if we assume that the coaches' decisions were generally *good* decisions (informed by years of experience), then a model that accurately predicts what the coach would prescribe effectively encodes the coaching staff's collective judgment about player readiness. When the model predicts high intensity, the player is in a state where an experienced coach would have prescribed a hard session.

Setup: We predicted Training Intensity Yesterday (continuous, [0,1)) from 15 morning covariates: Physical State, Mental State, Total Distance and High Speed Distance ACWR, Any ACWR Danger, Total Distance % and High Speed Distance % Yesterday, Perceived Exertion Yesterday, Total Minutes Yesterday, Avg Heart Rate and Heart Rate Exertion Yesterday, Days Since Game, Days Until Match, and Player ID. Four models were tested across lags 0–3. Dataset: 2,359 training, 634 test observations.

Results

Model	Best Lag	RMSE	R2	Pearson r
TabPFN	3	0.153	0.430	0.658
CatBoost	3	0.158	0.390	0.644
XGBoost	2	0.159	0.383	0.629
LinReg	3	0.194	0.080	0.396

The best model (**TabPFN, lag=3**) achieves **R2 = 0.43** and **Pearson r = 0.66**. This means 43% of the variance in the coaching staff's training intensity decisions is explained by observable morning-state variables.

Per-player breakdown: 20 of 22 test-set players have positive R2 (range 0.14 – 0.63). Only 2 players show negative R2 (–0.27 and –0.12). The model works consistently across the squad.

Feature importance (permutation importance, top 5):

Rank	Feature	Importance
1	Days Until Match	0.0102
2	Days Until Match (t-1)	0.0061
3	Total Distance % Yesterday	0.0039
4	Total Distance (m) Yesterday	0.0021
5	Total Distance % Yesterday (t-1)	0.0015

Match-cycle position (Days Until Match) is the strongest predictor, confirming that coaches follow a clear periodisation pattern.

Key Finding

The coaching staff's load decisions are systematic and largely driven by observable physiological signals (R2 = 0.43). The fitted model can serve as a continuous "Readiness to Train" score for each player-day. This is the primary actionable output of the project.

6. Experiment 3 — Status Decrease Prediction

As an auxiliary experiment, we investigated whether we can predict short-term player health outcomes. **Status Decrease** is a binary indicator capturing whether a player's medical status worsened from one day to the next (prevalence ~2%).

Setup: Logistic Regression, XGBoost, CatBoost, and TabPFN across 4 lags (0–3). Covariates include morning wellness, ACWR, GPS load history, Training Intensity Yesterday, schedule position, and playing position. Dataset: 3,780 training, 943 test observations (18 positive cases in test set).

Model	Best Lag	ROC-AUC	Precision	Recall	F1
CatBoost	2	0.640	0.022	0.444	0.043
LogReg	2	0.628	0.029	0.611	0.056
XGBoost	2	0.616	0.069	0.111	0.085
TabPFN	3	0.599	0.000	0.000	0.000

Status Decrease is extremely challenging to predict due to the severe class imbalance (~98% negative, ~2% positive, only 18 positive cases in the test set). The best model (CatBoost, AUC = 0.64) shows some discriminative ability above random (0.5) but with very low precision.

Conclusion

Status Decrease prediction shows modest signal (AUC = 0.64) but is limited by extreme class imbalance. The model could serve as a supplementary early-warning flag but should not be relied upon alone.

7. Conclusions & Recommendations

What we achieved

- **Data integration:** Merged four monitoring datasets into a unified, analysis-ready file with 46 features per player-day.
- **Experiment 1** confirmed that match-day performance is not predictable from training data (R^2 driven by Player ID; personal deviation R^2 near zero). This ruled out Match Intensity as a causal target.
- **Experiment 2** successfully recovered the coaching staff's load-assignment policy (best $R^2 = 0.43$, Pearson $r = 0.66$). The model serves as a data-driven Readiness to Train proxy, encoding the coaching staff's collective expertise.
- **Experiment 3** demonstrated modest ability to predict Status Decrease ($AUC = 0.64$), limited by extreme class imbalance.

Recommendations for OH Leuven

1. Deploy the Experiment 2 model as a **decision-support tool**. The predicted Training Intensity provides a data-driven "expected load" for each player-day. Deviations between model prediction and actual prescription flag unusual decisions.
2. Continue data collection. More seasons will improve model stability and enable validation across different squad compositions and fixture schedules.